

Project 2

INTERNAL EXPENSE AUDIT & FINANCIAL RISK DETECTION SYSTEM

Presented by Nurain Balqis

THE PROBLEM

Designed for a small services company without a dedicated accounting department, where physical receipts and expense claims were previously unmonitored.

- **Unstructured Data**
 - The original expense logs in [RAW DATA] were recorded in a non-standard database format, splitting entries manually across separate columns for each category. This made automated analysis or pivot-tabling impossible.
- **Lack of Financial Control**
 - There was no automated system to verify transaction validity, identify missing vendor details, or flag unusually high expenditures.

DATA CLEANING

ABC COMPANY (MARCH)					BARANG				KASARAN		
MINYAK					Bil.	Date	Place	Amount (RM)	Bil.	Date	Place
1	01/03/2026	CALTEX	RM	190.00	1	01/03/2026	IG	RM 14.00	1	01/03/2026	SEKUTUP
2	01/03/2026	PETRONAS	RM	20.00	2	02/03/2026	BARANG (HILTI)	RM 21.60	2	03/03/2026	SEKUTUP
3	01/03/2026	CALTEX	RM	10.00	3	02/03/2026	SEARIKAT P. MAGAIAJU	RM 24.50	3	04/03/2026	SEKUTUP
4	01/03/2026	CALTEX	RM	50.00	4	01/03/2026	IL SO	RM 112.00	4	09/03/2026	SEKUTUP
5	01/03/2026	PETRONAS	RM	100.00	5	02/03/2026	BARANG (HILTI)	RM 63.00	5	09/03/2026	SEKUTUP
6	01/03/2026	PETRONAS	RM	100.00	6	02/03/2026	IL (SOC)	RM 20.00	6	11/03/2026	SEKUTUP
7	01/03/2026	CALTEX	RM	100.00	7	02/03/2026	INSING L. DOND	RM 35.00	7	15/03/2026	SEKUTUP
8	01/03/2026	CALTEX	RM	100.00	8	02/03/2026	MUSCULAW ENTERPRISE	RM 26.00	8	17/03/2026	SEKUTUP
9	01/03/2026	PETRONAS	RM	200.00	9	04/03/2026	IL	RM 4.40	9	18/03/2026	SEKUTUP
10	01/03/2026	PETRONAS	RM	100.00	10	03/03/2026	IL	RM 508.00	10	31/03/2026	SEKUTUP
11	01/03/2026	CALTEX	RM	100.00	11	04/03/2026	IL	RM 387.00			
12	01/03/2026	CALTEX	RM	100.00	12	03/03/2026	IL	RM 139.00			
13	01/03/2026	PETRONAS	RM	200.00	13	05/03/2026	IL	RM 4.00			
14	01/03/2026	PETRONAS	RM	100.00	14	05/03/2026	IL	RM 18.00			
15	02/03/2026	PETRONAS	RM	200.00	15	05/03/2026	IL	RM 16.00			
16	02/03/2026	PETRONAS	RM	100.00	16	05/03/2026	IL	RM 27.00			
17	02/03/2026	CALTEX	RM	50.00	17	05/03/2026	IL	RM 34.60			
18	03/03/2026	PETRONAS	RM	100.00	18	05/03/2026	IL	RM 13.50			
19	03/03/2026	CALTEX	RM	100.00	19	07/03/2026	IL	RM 7.00			
20	03/03/2026	PETRONAS	RM	100.00	20	07/03/2026	IL	RM 208.00			
21	03/03/2026	PETRONAS	RM	100.00	21	04/03/2026	IL	RM 276.00			
22	02/03/2026	MINYAK	RM	100.00	22	05/03/2026	IL	RM 190.00			
23	03/03/2026	CALTEX	RM	165.00	23	04/03/2026	IL	RM 40.60			
24	04/03/2026	PETRONAS	RM	100.00	24	07/03/2026	IL	RM 348.00			
25	04/03/2026	CALTEX	RM	50.00	25	01/03/2026	IL	RM 2,245.00			
26	04/03/2026	SHELL	RM	150.00	26	07/03/2026	IL	RM 32.00			
27	04/03/2026	CALTEX	RM	100.00	27	05/03/2026	IL	RM 55.00			
28	02/03/2026	CALTEX	RM	200.00	28	05/03/2026	IL	RM 14.00			
29	04/03/2026	CALTEX	RM	200.00	29	07/03/2026	IL	RM 16.00			
30	04/03/2026	CALTEX	RM	100.00	30	08/03/2026	IL	RM 18.00			
31	05/03/2026	PETRONAS	RM	100.00	31	07/03/2026	IL	RM 278.20			
32	05/03/2026	CALTEX	RM	100.00	32	09/03/2026	IL	RM 1.90			
33	05/03/2026	PETRONAS	RM	150.00	33	09/03/2026	IL	RM 90.00			
34	05/03/2026	PETRONAS	RM	90.00	34	09/03/2026	IL	RM 30.00			
35	05/03/2026	PETRONAS	RM	10.00	35	10/03/2026	IL	RM 29.80			

RAW DATA

- Transformed a fragmented layout where expenses were separated manually into independent category blocks (Minyak, Barang, Kasaran) with their own repeating headers.
- Consolidated all scattered tables into a single, standardized, continuous tabular schema featuring explicit attributes: DATE, VENDOR, PRICE, CATEGORY, SOURCE, and DAY.
- Generated a DAY column directly from transaction dates to allow downstream analysis of weekly spending behaviors.
- Coded conditional logic to instantly separate standard transactions (Verified) from high-risk anomalies (Large Repair Check, High Refuel) and missing administrative details (Needs Vendor Detail)

AUDIT LOGIC & ANOMALY DETECTION

NO	DATE	VENDOR	PRICE	CATEGORY	SOURCE (O/B)	AUDIT STATUS	DAY
1	01/03/2026	CALTEX	RM 190.00	MINYAK	O	Verified	Sunday
2	01/03/2026	PETRONAS	RM 20.00	MINYAK	O	Verified	Sunday
3	01/03/2026	CALTEX	RM 10.00	MINYAK	O	Verified	Sunday
4	01/03/2026	CALTEX	RM 50.00	MINYAK	O	Verified	Sunday
5	01/03/2026	PETRONAS	RM 100.00	MINYAK	O	Verified	Sunday
6	01/03/2026	PETRONAS	RM 100.00	MINYAK	O	Verified	Sunday
7	01/03/2026	CALTEX	RM 100.00	MINYAK	O	Verified	Sunday
8	01/03/2026	CALTEX	RM 100.00	MINYAK	O	Verified	Sunday
9	01/03/2026	PETRONAS	RM 200.00	MINYAK	O	Verified	Sunday
10	01/03/2026	PETRONAS	RM 100.00	MINYAK	B	Verified	Sunday
11	01/03/2026	SUPLIER 1	RM 14.00	BARANG	O	Verified	Sunday
12	01/03/2026	SUPLIER 2	RM 112.00	BARANG	B	Verified	Sunday
13	01/03/2026	BARANG	RM 2,245.00	BARANG	B	Needs Vendor Detail	Sunday
14	01/03/2026	COMANY A	RM 441.00	KASARAN	O	Verified	Sunday
15	02/03/2026	PETRONAS	RM 200.00	MINYAK	O	Verified	Monday
16	02/03/2026	PETRONAS	RM 100.00	MINYAK	O	Verified	Monday
17	02/03/2026	CALTEX	RM 50.00	MINYAK	O	Verified	Monday
18	02/03/2026	MINYAK	RM 100.00	MINYAK	O	Verified	Monday
19	02/03/2026	CALTEX	RM 200.00	MINYAK	O	Verified	Monday
20	02/03/2026	CALTEX	RM 127.17	MINYAK	B	Verified	Monday
21	02/03/2026	BARANG	RM 21.60	BARANG	O	Needs Vendor Detail	Monday
22	02/03/2026	BARANG	RM 20.00	BARANG	O	Needs Vendor Detail	Monday
23	02/03/2026	SUPLIER 2	RM 35.00	BARANG	O	Verified	Monday
24	02/03/2026	SUPLIER 2	RM 26.00	BARANG	O	Verified	Monday
25	02/03/2026	SUPLIER 3	RM 24.50	BARANG	B	Verified	Monday
26	02/03/2026	SUPLIER 3	RM 63.00	BARANG	B	Verified	Monday
27	02/03/2026	WORKSHOP 1	RM 8.00	MAINTENANCE	O	Verified	Monday
28	02/03/2026	WORKSHOP 1	RM 24.00	MAINTENANCE	O	Verified	Monday
29	02/03/2026	WORKSHOP 2	RM 535.00	MAINTENANCE	B	Large Repair Check	Monday
30	02/03/2026	INFAK	RM 5,000.00	LAIN-LAIN	B	Verified	Monday
31	03/03/2026	PETRONAS	RM 100.00	MINYAK	O	Verified	Tuesday
32	03/03/2026	CALTEX	RM 100.00	MINYAK	O	Verified	Tuesday
33	03/03/2026	PETRONAS	RM 100.00	MINYAK	O	Verified	Tuesday
34	03/03/2026	PETRONAS	RM 100.00	MINYAK	O	Verified	Tuesday
35	03/03/2026	CALTEX	RM 165.00	MINYAK	O	Verified	Tuesday
36	03/03/2026	PETRONAS	RM 30.00	MINYAK	O	Verified	Tuesday
37	03/03/2026	PETRONAS	RM 150.00	MINYAK	O	Verified	Tuesday
38	03/03/2026	CALTEX	RM 190.00	MINYAK	B	Verified	Tuesday

CLEAN DATA

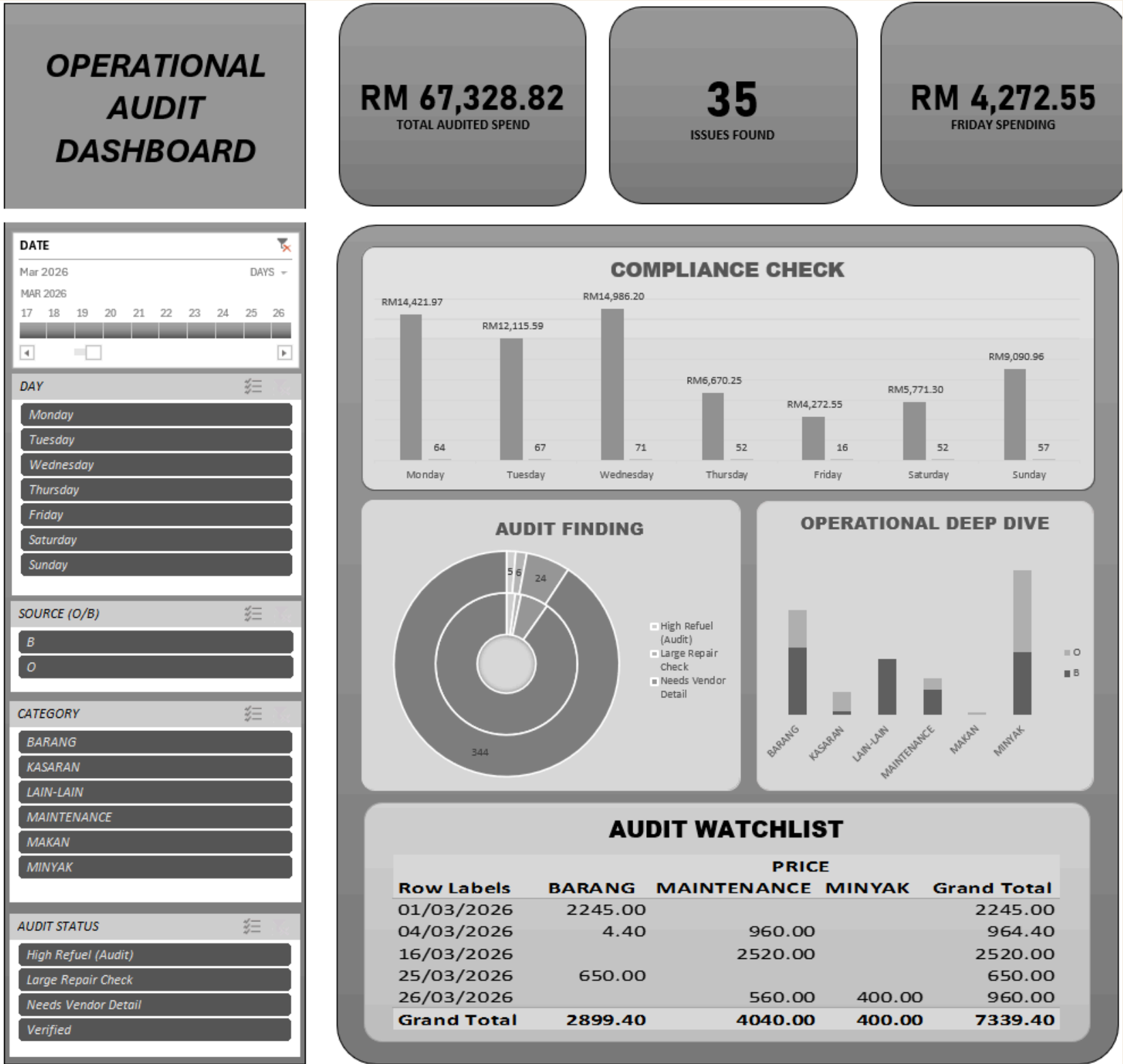
- Verified (344 Transactions): Cleared routine, compliant expenses backed by valid vendor data and complete receipts.
- Needs Vendor Detail (24 Transactions): Flagged incomplete entries lacking proper merchant names to preserve audit trails and secure tax compliance.
- Large Repair Check (6 Transactions): Isolated high-value maintenance costs, such as lorry and major machinery repairs, for immediate upper-management review.
- High Refuel Audit (5 Transactions): Programmed threshold alerts to catch abnormally high fuel claims deviating from standard vehicle fuel tank capacities.

DATA INSIGHT

PIVOT TABLE									
Sum of PRICE Row Labels Column Labels B O Grand Total					Row Labels Sum of PRICE Count of NO Monday Tuesday Wednesday Thursday Friday Saturday Sunday Grand Total				
BARANG	12416.05	6772.35	19188.40		RM 14,421.97	64			
KASARAN	791.90	3441.60	4233.50		RM 12,115.59	67			
LAIN-LAIN	10252.05		10252.05		RM 14,986.20	71			
MAINTENANCE	4668.00	1990.00	6658.00		RM 6,670.25	52			
MAKAN	116.50	432.40	548.90		RM 4,272.55	16			
MINYAK	11591.07	14856.90	26447.97		RM 5,771.30	52			
Grand Total	39835.57	27493.25	67328.82		RM 9,090.96	57			
AUDIT STATUS (Multiple Values)					Grand Total	RM 67,328.82	379		
Sum of PRICE Row Labels Column Labels BARANG MAINTENANC MINYAK Grand Total					Row Labels Sum of NO Count of NO2 High Refuel (Audit) Large Repair Check Needs Vendor Detail Verified Grand Total				
01/03/2026	2245.00		2245.00		1441	5			
04/03/2026	4.40	960.00	964.40		953	6			
16/03/2026		2520.00	2520.00		4617	24			
25/03/2026	650.00		650.00		64999	344			
26/03/2026		560.00	400.00	960.00	Grand Total	72010	379		
Grand Total	2899.40	4040.00	400.00	7339.40					

- Top Expense Driver: Fuel (MINYAK) emerged as the highest operational expense by frequency, commanding 224 out of 379 logged transactions.
- Supply Allocation: Project materials (BARANG) represented the second-largest operational cost chunk with 107 transactions.
- Weekly Trends: Highlighted spending peaks using the engineered DAY column to show the business owner exactly which days experienced the heaviest capital outflows.

DATA INSIGHT



- Executive KPI Cards: Summarizes for monitoring RM 67,328.82 in total audited spend and instantly flagging 35 issues requiring attention.
- Compliance Check (Weekly Trends): A chart mapping total expenditure alongside transaction frequency across the days of the week, helping isolate peak spending windows (e.g., Wednesdays hitting RM 14,986.20 over 71 transactions).
- Audit Finding & Deep Dive: A doughnut chart that breaks down risk severity (344 Verified vs. 35 Flags) paired with a stacked bar chart analyzing per-category funding sources (O vs. B).
- Actionable Audit Watchlist: Pinpoints the exact dates and cost categories (such as BARANG, MAINTENANCE, and MINYAK) where high-risk anomalies occurred, giving management an instant roadmap for review.

INTERACTIVE USER INTERFACE (UI) FOR BUSINESS STAKEHOLDERS

BUSINESS VALUE & IMPACT

- Financial Leakage Prevention: Stopped unchecked spending by instantly exposing high-risk anomalies and missing receipts.
- Tax & Audit Readiness: Transformed a messy spreadsheet into a fully auditable database, ensuring the company is completely prepared for official tax filings.
- Data-Driven Governance: Provided the business owner with a framework to step away from manual receipt checking and manage company cash flow through high-level dashboard exceptions.



2026

THANK YOU

Thanks For Watching



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